

Bayesian Hill Mixture Models for Marketing Mix Modeling: Synthetic Benchmarks, Component Resolvability, and Real-Data Evaluation

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Abstract

Marketing Mix Models (MMMs) typically use a single Hill saturation curve per channel. We study a Bayesian *predictive mixture* of K Hill curves, implemented in NumPyro with an ordering reparameterisation, post-hoc relabeling, and label-invariant convergence diagnostics. The model is a mixture over the observation likelihood rather than over latent consumer segments: at each time step the aggregate target y_t is modelled as a Gaussian mixture whose components correspond to different Hill responses on the same adstocked spend. We make no individual-level claim from this structure. All experiments—synthetic and real—use a single aggregated spend proxy $x_t = \sum_c x_{c,t}$ rather than channel-level mixtures, so the empirical scope of the paper is a controlled one-dimensional version of MMM rather than a full channel-level deployment. The question we ask is whether the extra component capacity buys better aggregate predictive density on this controlled scope and recovers interpretable response curves.

We report three experiments. (i) A controlled synthetic benchmark shows that, when the data-generating process is itself a Hill mixture, the model recovers it: estimated ELPD-LOO improves whenever the DGP has multiple Hill components and incurs only a small penalty when it does not. (ii) A *component resolvability* study varies the cosine distance between true Hill curves and identifies an empirical, model-dependent transition near $\bar{d}_{\cos} \approx 0.10$, below which posteriors tend to collapse to fewer effective components. (iii) A real-data benchmark on three Conjura organisations across three verticals (beauty & fitness, home & garden, toys & hobbies) shows that Mixture $K=3$ has higher estimated ELPD-LOO than single-Hill in every organisation, by +222 to +308 units; these gaps are reported subject to a paired `se_diff` that we could not recompute from the stored summaries and to Pareto- k reliability flags on Beauty & Fitness. The gain is concentrated in predictive density: holdout MAPE and CRPS do not improve uniformly, and on Beauty & Fitness they deteriorate slightly under $K = 3$.

The benchmark convergence threshold ($\hat{R} \leq 1.05$) that 44 of 45 synthetic fits satisfy fails uniformly for Mixture $K=3$ on real data, where every fit reports $\max \hat{R} \geq 1.05$ and several show severe chain mixing failures ($\max \hat{R}$ up to 3.6, tail ESS per chain as low as 1–6). Re-using the resolvability axis on the real posteriors shows that this gap is organisation-dependent: Home & Garden sits at the resolvability boundary, Toys & Hobbies is consistent with three resolvable components but with chain-mixing caveats, and Beauty & Fitness shows strong multi-modality in the posterior coupled with influential observations. We report these three regimes directly rather than collapsing them into a single summary. Code, configurations, and per-fit artifacts are released for reproduction.

Keywords: Marketing Mix Modeling, Bayesian inference, Hill function, mixture models, label switching, component resolvability

1 Introduction

Marketing Mix Modeling (MMM) enables organisations to quantify the effectiveness of advertising spend and to optimise budget allocation. The canonical formulation fits a single response curve per channel on aggregate time-series data, combining geometric (or Weibull) adstock for carryover with a Hill function for saturation [Jin et al., 2017, Chan and Perry, 2017]. We refer to this baseline as *Standard MMM*.

Standard MMM commits to a single saturation curve per channel. There is no observational guarantee that aggregate spend–response data are well described by exactly one Hill curve: nonlinearities can arise from regime shifts, time-varying mixing weights across audiences, or simple model misspecification, and an aggregate single-Hill fit can absorb those into a biased average curve. This paper studies a Bayesian *predictive mixture* of Hill curves as a minimal capacity extension of Standard MMM, and asks two operational questions: how much component separation must be present in the data-generating process before the mixture posterior can recover it, and how well do the recovery guarantees that hold on synthetic data transfer to field data? The empirical scope is deliberately narrow: we collapse all channels into a single aggregated spend series $x_t = \sum_c x_{c,t}$ and study mixture-over-saturation behaviour on that one-dimensional problem. A channel-level mixture-of-Hill (with mixing weights and Hill components within each channel) is a direct extension but is not what we evaluate here, and we flag the cases where the single-aggregate framing materially restricts the conclusions. We deliberately do not claim that the recovered mixture components correspond to underlying consumer segments. The model is fit to aggregate daily time series and cannot identify individual-level structure; we use “component” throughout in the sense of “predictive mixture component”, and discuss this scope choice explicitly in the limitations.

Contributions.

1. A controlled synthetic benchmark comparing single-Hill and Bayesian mixture-of-Hill models under known ground truth, with predictive, latent-recovery, and convergence metrics.
2. A *component resolvability* study that systematically varies the cosine distance between true Hill curves and traces the recovered Shannon effective component count as a function of true separation, identifying an empirical threshold below which the posterior collapses to fewer components.
3. A real-data benchmark on three Conjura organisations across three verticals, paired with a re-use of the resolvability axis on the real posteriors. The combination shows that mixture-of-Hill has higher estimated predictive density than single-Hill across organisations (subject to PSIS reliability and paired-SE caveats) but that publication-ready convergence is organisation-dependent in a way the resolvability framework can diagnose.

2 Related Work

Bayesian MMM with adstock and saturation. The combination of geometric adstock with Hill saturation under a Bayesian likelihood is the canonical modern MMM formulation [Jin et al., 2017, Chan and Perry, 2017]. Subsequent open-source implementations (PYMC-MARKETING, MERIDIAN) have made this family practical at industrial scale; we keep our model in the same family so that the mixture extension is a minimal, drop-in change rather than a separate architecture. We deliberately do not extend the dynamics: Dew et al. [2024] argue that nonlinear and time-varying effects in MMM are jointly not identified from typical observational time series, so apparent

saturation can be an artefact of misspecified dynamics. Their critique applies to our setting too; we do not attempt to resolve it. We instead fix the dynamics to standard geometric adstock and study what happens when only the saturation component is generalised from a single Hill curve to a Hill mixture.

Heterogeneous response and segmentation. The marketing literature on consumer heterogeneity [Wedel and Kamakura, 2000, Allenby and Rossi, 1998] works at the individual or panel level, where latent-class and hierarchical Bayes models can be identified against the data. MMM operates on aggregated daily time series and *cannot* resolve consumer-level structure from such data; in this sense the mixture-of-Hill posterior is not directly comparable to those approaches. What it offers is a strictly larger predictive model class within the standard MMM family, so that aggregate departures from a single Hill curve do not have to be silently absorbed into a biased single-curve fit.

Mixture identifiability and label switching. Finite-mixture posteriors are non-identified under permutation of component labels [Stephens, 2000, Frühwirth-Schnatter, 2006]. The label-switching problem makes per-component summaries meaningless without an additional convention, and breaks standard split- $\hat{R}$ diagnostics [Vehtari et al., 2021]. The two standard remedies—ordering constraints and post-hoc relabeling [Papastamoulis, 2014]—trade off detailed balance against interpretability. We combine both with an additional label-invariant diagnostic (Section 3.1) and treat them as complementary rather than substitutable.

3 Model Specification

Let x_t denote aggregated marketing spend at time t and y_t the observed outcome (purchases, revenue, or a related target). Figure 1 shows the model in plate notation. The model is implemented in NumPyro [Phan et al., 2019] with JAX acceleration.

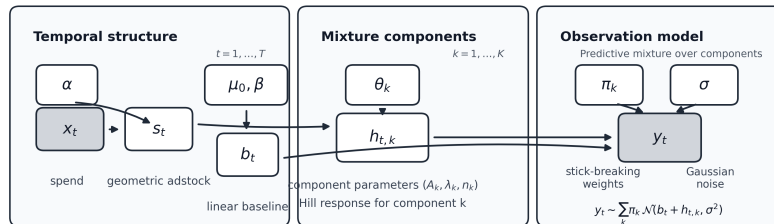


Figure 1: Graphical model. Observed variables are shaded; plates indicate replication over time steps t and mixture components k .

Adstock transformation. Geometric decay captures carryover from previous periods:

$$s_t = x_t + \alpha \cdot s_{t-1}, \quad s_0 = 0, \quad \alpha \sim \text{Beta}(2, 2). \quad (1)$$

Hill saturation. Each mixture component $k \in \{1, \dots, K\}$ has its own Hill response:

$$f_k(s) = A_k \cdot \frac{s^{n_k}}{\lambda_k^{n_k} + s^{n_k}}, \quad (2)$$

with maximum effect A_k , half-saturation point λ_k , and slope n_k (priors specified in the next paragraph and fully detailed in Appendix B).

Mixture likelihood. The observation model is a Gaussian mixture over components:

$$y_t \sim \sum_{k=1}^K \pi_k \cdot \mathcal{N}(\mu_0 + \beta t + f_k(s_t), \sigma^2), \quad (3)$$

where μ_0, β capture intercept and trend, and $\pi = (\pi_1, \dots, \pi_K)$ is constructed by a stick-breaking parameterisation $\pi_k = v_k \prod_{j < k} (1 - v_j)$ with $v_k \sim \text{Beta}(a, b)$ (see Appendix B for the exact (a, b) used).

Hierarchical, data-anchored priors. To accommodate diverse data scales we use mildly hierarchical, data-anchored priors rather than a single LogNormal on (A_k, n_k) : per-component $(\log A_k, \log n_k)$ are drawn from a Normal with shared location $\mu_{\log A}, \mu_{\log n}$ and component-wise scales $\sigma_{\log A}, \sigma_{\log n}$, with $\mu_{\log A}$ centred on $\log(0.3 \cdot \text{range}(y))$. The intercept and noise priors $\mu_0 \sim \mathcal{N}(\bar{y}, 2\sigma_y)$, $\sigma \sim \text{HalfNormal}(\sigma_y)$ are data-anchored in the same spirit. The full prior specification, including all hyperparameters, is in Appendix B.

3.1 Identifiability and Label Switching

A finite mixture is invariant under permutations of component labels: any reordering of $(\pi_k, A_k, \lambda_k, n_k)$ yields the same likelihood. As a result, posterior summaries computed per component are meaningless without an additional convention, and standard split- $\hat{R}$ on raw component parameters cannot distinguish genuine non-mixing from label permutations across chains [Stephens, 2000, Frühwirth-Schnatter, 2006].

Within-MCMC ordering with data anchors. The reported fits impose $\lambda_1 < \lambda_2 < \dots < \lambda_K$ during sampling via an ordered reparameterisation: $\log \lambda_1$ is anchored at a low quantile of the observed adstock $\hat{\lambda}_1 = \text{quantile}_{q_1}(s)$ and the remaining $\log \lambda_k$ are obtained by adding strictly positive increments around anchor gaps derived from higher quantiles $q_2, \dots, q_K$ of s (Appendix B). This restricts the posterior to one canonical labelling and stabilises mixing relative to a flat ordered-LogNormal prior, at the cost of mild posterior geometry distortion near the anchors.

Post-hoc relabeling for summaries. Per-component posterior summaries are additionally reported after a post-hoc reordering of each draw to $\lambda_1 < \dots < \lambda_K$ [Stephens, 2000, Papastamoulis, 2014]. The within-MCMC ordering and post-hoc relabeling are not used as competing strategies; the in-sampler ordering is the primary identification mechanism, and post-hoc relabeling is applied to the resulting draws when computing the component-wise diagnostics below.

Label-invariant diagnostics. We compute $\hat{R}$ and effective sample sizes on three groups of quantities [Vehtari et al., 2021]:

1. *Joint log-likelihood* $\sum_t \log \sum_k \pi_k \phi(y_t | \mu_t^{(k)}, \sigma)$, which is label-invariant by construction.
2. *Scalar parameters* $(\alpha, \sigma, \mu_0, \beta)$ that are not affected by label permutations.
3. *Relabeled component parameters*, after the post-hoc reordering above.

A model is treated as effectively converged when all three pass; this is the criterion we use to report a *publication-ready* fit. Concretely, a fit is publication-ready when (i) the No-U-Turn sampler produces zero divergences, zero maximum-tree-depth saturations, and a minimum chain-level BFMI of at least 0.30; and (ii) every retained mixing diagnostic satisfies $\hat{R} \leq 1.01$ with both bulk and tail effective sample size of at least 100 per chain. The mixing diagnostics retained depend on the model: for single-Hill we use raw parameter $\hat{R}/\text{ESS}$; for mixtures we replace the raw component-parameter diagnostics with the label-invariant diagnostics above and additionally require the relabeled component parameters to satisfy the same thresholds. These thresholds match the conservative defaults of Vehtari et al. [2021] and ensure that “pass” is a strict criterion rather than a soft one.

4 Experimental Setup

Inference. All fits use NUTS with two chains, `init_to_median`, diagonal mass adaptation, and model-dependent warmup/sampling lengths and target acceptance (full schedule in Appendix A).

Data splits. Each time series is divided into a 75/25 train/test split. Holdout predictions inherit the adstock state accumulated during training and are evaluated at their absolute time index rather than reset to zero, so that adstock decay and the trend term are propagated forward consistently.

Evaluation metrics. We report (i) expected log pointwise predictive density estimated by Pareto-smoothed importance-sampling leave-one-out cross-validation (ELPD-LOO), with the count of $\hat{k}_i > 0.7$ observations reported alongside as a reliability flag; (ii) test-set mean absolute percentage error (MAPE), continuous ranked probability score (CRPS), and 90% predictive coverage; and (iii) for synthetic conditions, latent-mean normalised RMSE against the known DGP latent.

Publication-ready convergence. We apply the publication-ready criterion of Section 3.1 unchanged across all experiments.

Component summaries. For mixture fits we additionally report posterior mixture weights $\hat{\pi}_k$, the Shannon effective component count $^1D = \exp(-\sum_k \hat{\pi}_k \log \hat{\pi}_k)$ [Hill, 1973], and the mean pairwise cosine distance between active Hill response curves. The latter is computed on a fit-adaptive saturation grid that covers the active components’ half-saturation range, making it independent of how the prior scale was set.

5 Synthetic Benchmark

5.1 Data-Generating Processes

We evaluate three data-generating processes (DGPs)—*Single* ($K = 1$), *Mixture* $K = 2$, *Mixture* $K = 3$ —against three fitted models—Single Hill, Mixture $K = 2$, Mixture $K = 3$. Each DGP

uses $T = 200$ observations with five random seeds, so the design contains $3 \text{ DGPs} \times 3 \text{ models} \times 5 \text{ seeds} = 45 \text{ fits}$.

5.2 Results

Convergence. Under the benchmark threshold $\hat{R} \leq 1.05$ (raw parameters for single-Hill, label-invariant diagnostics for mixtures), 44/45 fits cleared (Figure 2); the single exception was Mixture $K = 3$ on the $K = 2$ DGP at seed 1, with $\max \hat{R} \approx 1.5$. The strict publication-ready criterion ($\hat{R} \leq 1.01$, ESS $\geq 100/\text{chain}$, plus HMC clauses of Section 3.1) is met by 33/45; the rest fall in the warning band $\hat{R} \in [1.01, 1.05]$. Synthetic plots use the loose threshold for comparability with published MMM benchmarks; real-data results in Section 7 use the strict criterion.

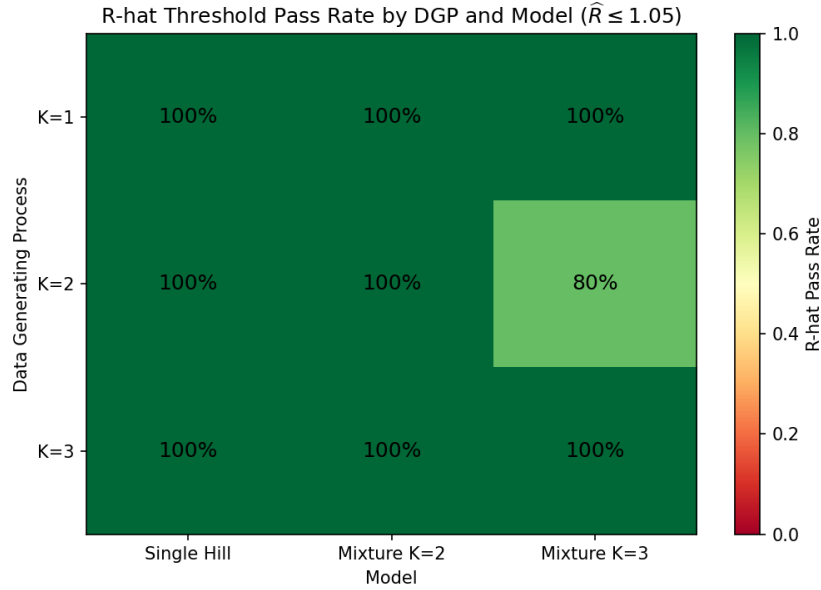


Figure 2: Benchmark convergence pass rate ($\hat{R} \leq 1.05$ on the retained mixing diagnostics) by DGP and model on the synthetic benchmark. Eight cells reach 100%; the Mixture $K = 3$ model on the $K = 2$ DGP reaches 80% (4 of 5 seeds), with seed 1 failing at $\max \hat{R} \approx 1.5$.

Predictive performance. Figures 3 and 4 report ELPD-LOO and holdout CRPS. When the true DGP is homogeneous ($K = 1$), the three models tie within standard error. When heterogeneity is present, mixtures gain substantially in ELPD ($\Delta \approx +50$ on $K = 2$, $\Delta \approx +45$ on $K = 3$). CRPS gains are noticeably smaller than ELPD gains ($\Delta \approx 0.1$ on $K = 2$, $\Delta \approx 0.4$ on $K = 3$), indicating that the bulk of the improvement is in probabilistic density rather than in point prediction.

Latent recovery. Figure 5 shows latent-mean nRMSE on the test split. The pattern mirrors ELPD: mixtures roughly halve nRMSE under $K = 2$ ($0.14 \rightarrow 0.07$) and reduce it by a factor of ≈ 2.4 under $K = 3$ when the model is matched to the true K ($0.085 \rightarrow 0.035$). This shows that the ELPD gains correspond to genuinely better recovery of the latent response, not just better marginal calibration.

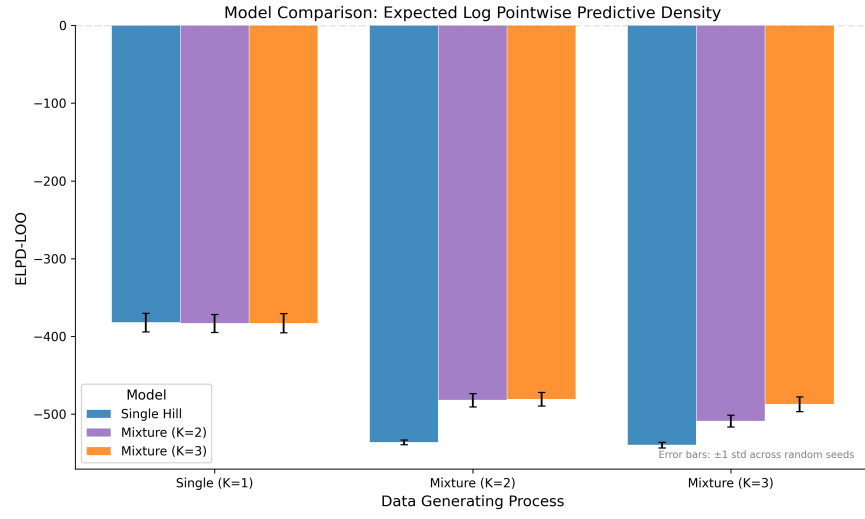


Figure 3: ELPD-LOO by DGP and model on the synthetic benchmark. Higher is better. Mixtures improve substantially when $K \geq 2$.

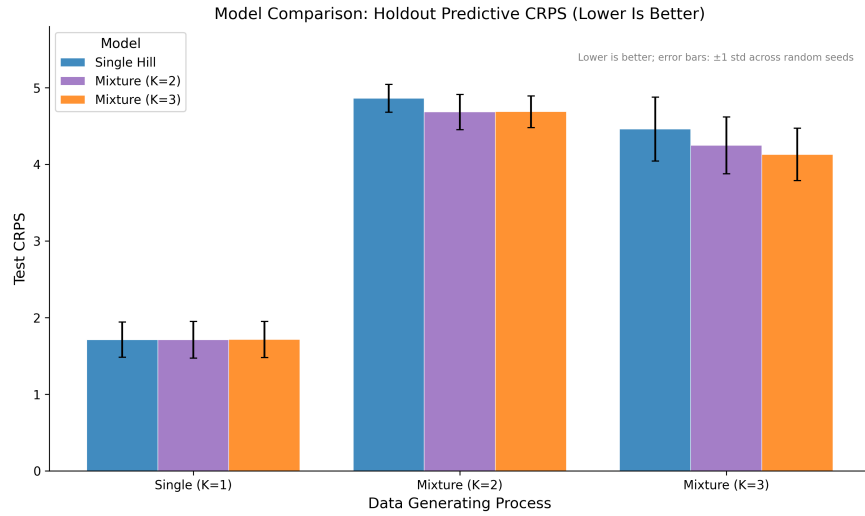


Figure 4: Holdout CRPS by DGP and model on the synthetic benchmark. Lower is better. Gains are smaller in magnitude than ELPD gains.

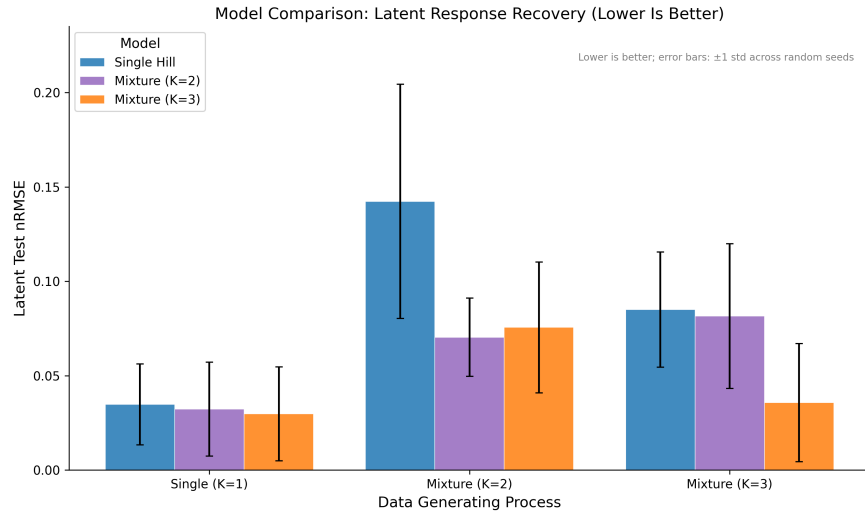


Figure 5: Latent-mean nRMSE by DGP and model. Lower is better. Mixtures recover the latent response more accurately when heterogeneity is present.

6 Component Resolvability

The previous section shows that mixtures help when heterogeneity exists, but does not tell us *how much* heterogeneity is required for the posterior to actually separate components. We address this with a controlled study.

6.1 Design

We construct nine DGP profiles by varying the spacing δ between true Hill half-points around a fixed centre c , holding amplitude and slope fixed. For $K_{\text{true}} = 1$ we use one anchor profile; for $K_{\text{true}} \in \{2, 3\}$ we use four profiles each, spanning low to extreme separation. Each profile is run with five seeds and fit with both Mixture $K = 2$ and Mixture $K = 3$, for $9 \times 2 \times 5 = 90$ fits in total. The separation axis $\bar{d}_{\text{cos}}$ and recovery axis 1D are the cosine-separation and Shannon-effective-count measures defined in Section 4; here $\bar{d}_{\text{cos}}$ is computed on the *true* component curves and 1D on the posterior mixture weights.

6.2 Results

Figure 6 plots 1D against $\bar{d}_{\text{cos}}$ across all 90 fits. Three regimes are visible:

- For $\bar{d}_{\text{cos}} \lesssim 0.05$, both models collapse toward the trivial value: even when $K_{\text{true}} \geq 2$, the posterior cannot resolve more than ≈ 1.5 – 1.7 effective components.
- Between $\bar{d}_{\text{cos}} \approx 0.05$ and 0.3 , the recovered count rises sharply with separation: this is the resolvable transition region.
- For $\bar{d}_{\text{cos}} \gtrsim 0.3$, the recovered count saturates near K_{true} . Mixture $K = 2$ saturates near 2 by construction; Mixture $K = 3$ tracks the true K and approaches 3 on $K = 3$ DGPs.

Failure modes. The publication-ready convergence criterion fails most frequently in two regimes. First, under-specified fits at low separation: Mixture $K = 2$ on $K_{\text{true}} = 3$ DGPs at low δ shows publication-fail rates of 0.6–0.8. Second, at the highest separation under $K_{\text{true}} = 3$, Mixture $K = 3$ exhibits a non-trivial failure rate (0.6 at the most extreme profile), suggesting that the strict diagnostic criterion is sensitive to posterior geometry even when the components are far apart.

Practical implication. The cosine ≈ 0.1 figure is an empirical, model-dependent transition, not a sharp threshold: e.g. Mixture $K = 3$ on $K_{\text{true}} = 3$ already recovers Shannon count ≈ 2.27 at true cosine 0.058. The qualitative picture is that separations below ≈ 0.05 are reliably under-resolved and separations above ≈ 0.3 are reliably recovered when K matches, with sensitivity to model order and seed in between. We use these regimes as a vocabulary for Section 7.4, not as a deployable rule.

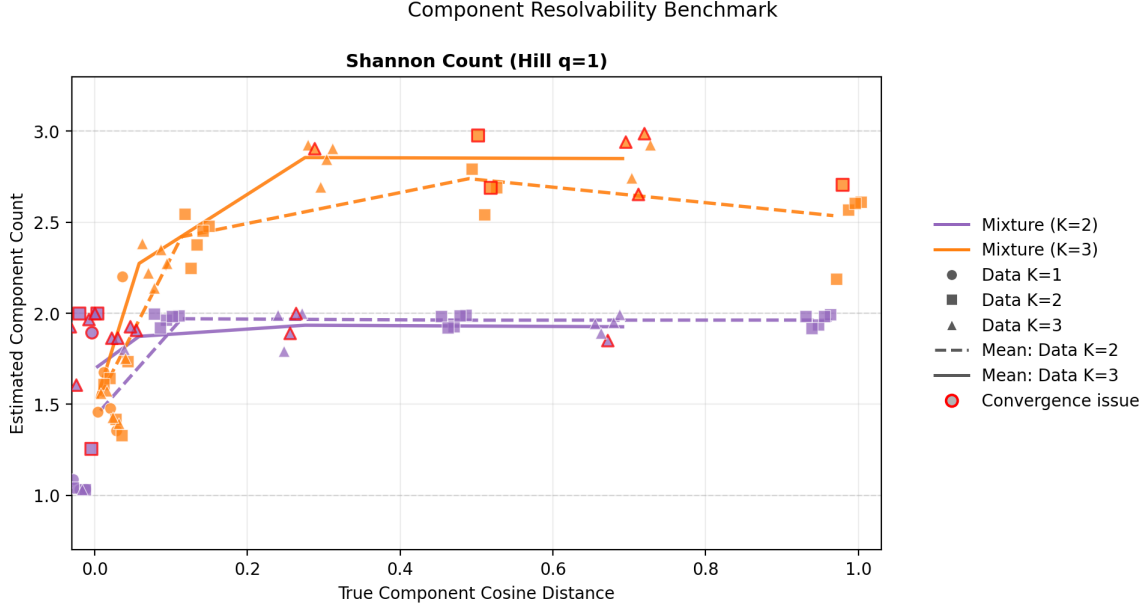


Figure 6: Component resolvability: Shannon effective count of the posterior versus mean pairwise cosine distance of the true Hill curves. Red-outlined points indicate cases that failed the publication-ready convergence criterion.

7 Real-Data Benchmark

7.1 Dataset and Organisation Selection

We use the public Conjura MMM dataset, which contains daily spend across nine advertising channels and several purchase-related target variables for a panel of e-commerce organisations. Following the per-organisation modelling convention of MMM practice, we treat each organisation as an independent time series and do not share parameters across organisations.

To select organisations for the benchmark, we screen all candidates with at least 200 daily observations and at least two active spend channels (non-zero on more than 10% of days), and rank them by a composite score combining series length, spend coefficient of variation, spend–target correlation, and channel breadth. We pick three top-ranked organisations from three different verticals to support a modest cross-vertical claim. The selected organisations are summarised in Table 1.

Table 1: Selected Conjura organisations for the real-data benchmark.

Vertical	T (days)	Active channels	Spend–target r	Total spend (USD M)
Toys & Hobbies	1,375	5	0.95	3.3
Beauty & Fitness	924	8	0.96	5.2
Home & Garden	1,642	5	0.80	2.1

7.2 Predictive Performance

Each organisation is fit with single-Hill, Mixture $K = 2$, and Mixture $K = 3$ under three seeds, giving 27 fits in total; inference and evaluation follow Section 4.

Table 2 reports ELPD-LOO, holdout MAPE, holdout CRPS, and holdout 90% coverage averaged across seeds. The key observation is that *Mixture $K = 3$ has the highest estimated ELPD-LOO in every organisation*, with point gaps of +222 (beauty & fitness), +308 (home & garden), and +275 (toys & hobbies) over the single-Hill baseline. Mixture $K = 2$ is intermediate. Both the magnitude and the ranking should be read alongside the paired-SE and PSIS reliability caveats below; we report them as estimates rather than as established improvements.

Table 2: Real-data predictive performance, averaged across three seeds. ELPD-LOO is reported with the per-fit PSIS-LOO standard error in parentheses (averaged across seeds); higher is better. MAPE/CRPS lower is better; 90% coverage closer to 0.9 is better. Bold marks the best ELPD-LOO per organisation.

Organisation	Model	ELPD-LOO ($\pm$ SE)	Test MAPE	Test CRPS	90% cov.
Beauty & Fitness	Single Hill	$-3,008 \pm 86$	30.3	108	0.33
	Mixture $K = 2$	$-2,992 \pm 96$	29.6	101	0.37
	Mixture $K = 3$	$-2,786 \pm 48$	34.3	122	0.54
Home & Garden	Single Hill	$-6,168 \pm 63$	26.0	25.0	0.87
	Mixture $K = 2$	$-5,985 \pm 56$	31.4	27.2	0.88
	Mixture $K = 3$	$-5,860 \pm 48$	28.2	25.9	0.86
Toys & Hobbies	Single Hill	$-6,379 \pm 89$	36.4	57.3	0.94
	Mixture $K = 2$	$-6,346 \pm 89$	29.3	53.5	0.98
	Mixture $K = 3$	$-6,104 \pm 91$	34.3	47.3	0.96

Two caveats apply to the magnitude of these gaps before any substantive reading.

Paired SE. The standard errors in Table 2 are per-fit PSIS-LOO standard errors of the individual ELPD point estimates, not the proper paired difference `se.diff` between two models on the same observations [Vehtari et al., 2017]. Computing `se.diff` requires pointwise log-likelihood matrices that we did not retain. As a coarse heuristic that is conservative only if the two models’ pointwise scores are positively correlated, treating the two model ELPDs as independent gives $\sqrt{86^2 + 48^2} \approx 99$ (beauty & fitness), $\sqrt{63^2 + 48^2} \approx 79$ (home & garden), and $\sqrt{89^2 + 91^2} \approx 127$ (toys & hobbies); the $K = 3$ vs. single-Hill gaps are then roughly 2–4 such heuristic SE in magnitude. This is not a paired `se.diff` substitute—the direction of the bias depends on the (unknown) pointwise covariance—so we report it only to ground readers in the order of magnitude. The seed-level SDs in Figure 7 are Monte Carlo variability across NUTS reruns, not data-level uncertainty.

PSIS reliability. PSIS-LOO itself is reliable only when $\hat{k}_i \leq 0.7$ for essentially all observations. Table 3 reports the per-cell $\hat{k}_i > 0.7$ counts: Beauty & Fitness Mixture $K = 2$ has 15 such observations on average (max 21), Beauty & Fitness Mixture $K = 3$ has 3.3 (max 10), and other cells are in the single digits. The ELPD-LOO numbers for Beauty & Fitness in particular should therefore be read as ELPD estimates from a partially-degraded PSIS approximation, not as clean predictive-density measurements; the relative ordering of models is informative but the absolute magnitudes are softer.

Figure 7 visualises the same comparison. The Mixture $K = 3$ advantage is consistent in sign across organisations, while the absolute scale of ELPD-LOO varies by an order of magnitude with

series length and target variance.

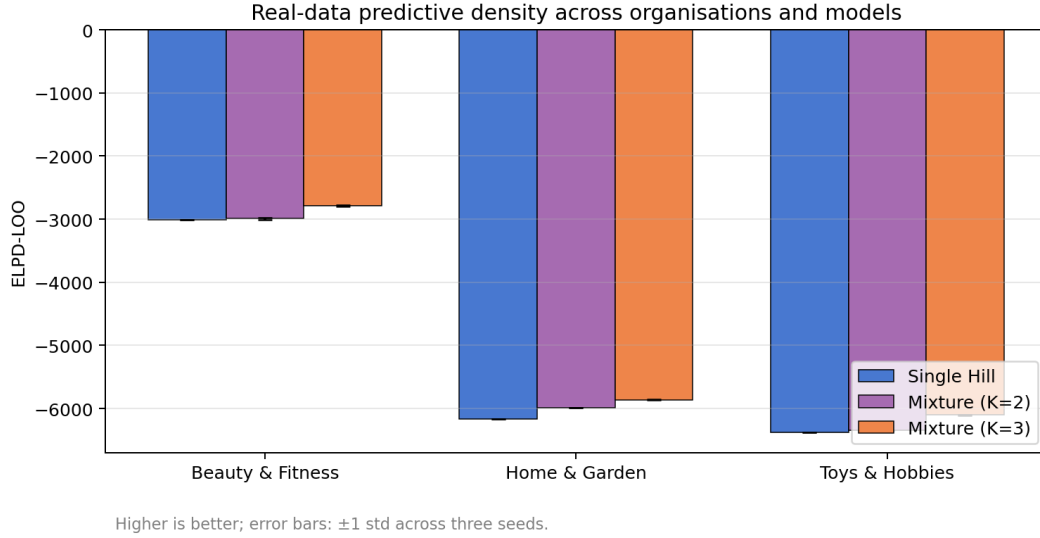


Figure 7: Real-data ELPD-LOO across the three organisations and three models. Higher is better. Error bars are ± 1 standard deviation across three NUTS reruns at different random seeds and represent Monte Carlo variability of the sampler, not data-level uncertainty; in particular they should not be read as paired `se_diff` between models. Mixture $K = 3$ has the highest estimated mean ELPD-LOO in every organisation, subject to the paired-SE and PSIS reliability caveats discussed after Table 2.

The relationship between ELPD and the other metrics is heterogeneous and *not* uniformly favourable to the mixture. On *home & garden*, Mixture $K = 3$ has the best ELPD but Single Hill has the best MAPE (26.0) and CRPS (25.0); the mixture trades a small MAPE/CRPS regression for a sizeable ELPD gain. On *toys & hobbies*, ELPD and CRPS both improve under $K = 3$, while MAPE is non-monotonic. On *beauty & fitness*, ELPD improves substantially but MAPE and CRPS *deteriorate* for $K = 3$, and the model remains visibly underconfident: 90% coverage rises from 0.33 (single) to 0.54 ($K = 3$) but stays well below the nominal level. The ELPD gain on this organisation therefore reflects a localised improvement in predictive density rather than a uniform predictive win; this should not be read as “ $K = 3$ predicts better on beauty & fitness”. We return to this in the discussion.

To make concrete what the mixture is doing, Figure 8 shows the posterior response curves for the home & garden organisation under all three models on a single representative seed. Single-Hill recovers one slowly-saturating curve. Mixture $K = 2$ resolves two components with weights 0.93 and 0.07, in which the small minority component carries a much higher saturation ceiling. Mixture $K = 3$ uses two active components with weights 0.84 and 0.14 and a third component that collapses to weight 0.02 (labelled *inactive* in the legend). The qualitative two-mode structure is robust to whether K is set to 2 or 3, which is consistent with the posterior effective component count reported below.

7.3 Convergence on Real Data

Table 3 summarises the rate at which each (organisation, model) cell passes the publication-ready convergence criterion (HMC diagnostics plus label-invariant and relabeled $\hat{R}$) across the three seeds.

Posterior response curves — Home & Garden (seed 0)

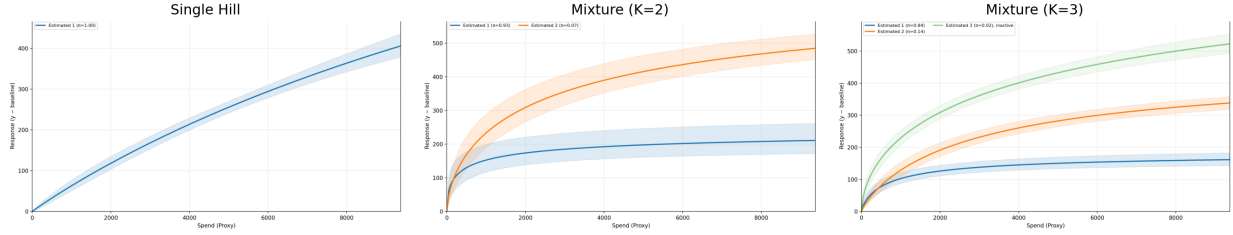


Figure 8: Posterior response curves for Home & Garden (seed 0) under Single Hill, Mixture $K = 2$, and Mixture $K = 3$. Shaded bands are 90% posterior intervals; legend entries report posterior mixture weights $\hat{\pi}$ per component. The Mixture $K = 3$ third component is labelled *inactive* because its weight collapses to 0.02.

Two patterns are stark.

Table 3: Per-cell convergence and component-summary diagnostics for the real-data benchmark. Columns are defined in Sections 4 and 7.4.

Organisation	Model	Pub. pass	$^1D_{\text{post}}$	K_{act}	$\hat{k} > 0.7$ (mean/max)	$\bar{d}_{\text{cos}}^{\text{post}}$
Beauty & Fitness	Single Hill	0.67	1.00	1.00	1.7 / 3	—
	Mixture $K = 2$	0.00	1.26	1.65	15.0 / 21	0.46
	Mixture $K = 3$	0.00	1.79	2.17	3.3 / 10	0.68
Home & Garden	Single Hill	0.67	1.00	1.00	0.7 / 1	—
	Mixture $K = 2$	0.33	1.27	1.94	0.7 / 1	0.09
	Mixture $K = 3$	0.00	1.66	2.00	1.7 / 3	0.12
Toys & Hobbies	Single Hill	1.00	1.00	1.00	2.7 / 4	—
	Mixture $K = 2$	0.67	1.50	2.00	1.7 / 2	0.47
	Mixture $K = 3$	0.00	2.72	2.45	3.3 / 4	0.29

First, *Mixture $K = 3$ passes the publication-ready criterion in 0/9 fits* on real data, despite the matched- K synthetic fits passing in 5/5 runs at matched $K = 3$ (Section 5). The Mixture $K = 3$ failures on real data are also not all the same kind: Home & Garden chains hit the strict threshold only marginally, whereas Beauty & Fitness and Toys & Hobbies report max $\hat{R}$ of 2.4–3.6 and 1.8, with tail ESS per chain of 1–6 and 23–34 respectively, and large numbers of max-tree-depth saturations. We therefore treat the three organisations’ $K = 3$ fits as qualitatively different failure modes rather than a single “0/9” bucket (Section 7.4). Second, the Shannon effective count $^1D_{\text{post}}$ is at most 2.72 (Toys & Hobbies $K = 3$), while in two of the three organisations it stays below 1.8, suggesting that the draws rarely use the full $K = 3$ capacity in those cases.

The high Pareto- k count for the beauty & fitness Mixture $K = 2$ fit (15 observations on average) is consistent with a moderately misspecified model on data with influential outliers, and points to robustifying the likelihood as a useful next step.

7.4 Where Real Organisations Sit on the Resolvability Axis

The synthetic resolvability study (Section 6) characterises recovery as a function of *true* cosine separation between component curves. Real data has no ground truth, but we can compute the same quantity from each fit’s *posterior* component summary: the mean pairwise cosine distance between

the active Hill curves implied by the posterior means (A_k, λ_k, n_k) , on a fit-adaptive saturation grid. Plotting the recovered Shannon effective count against this posterior separation places the real organisations onto the same plane as the synthetic resolvability fits (Figure 9).

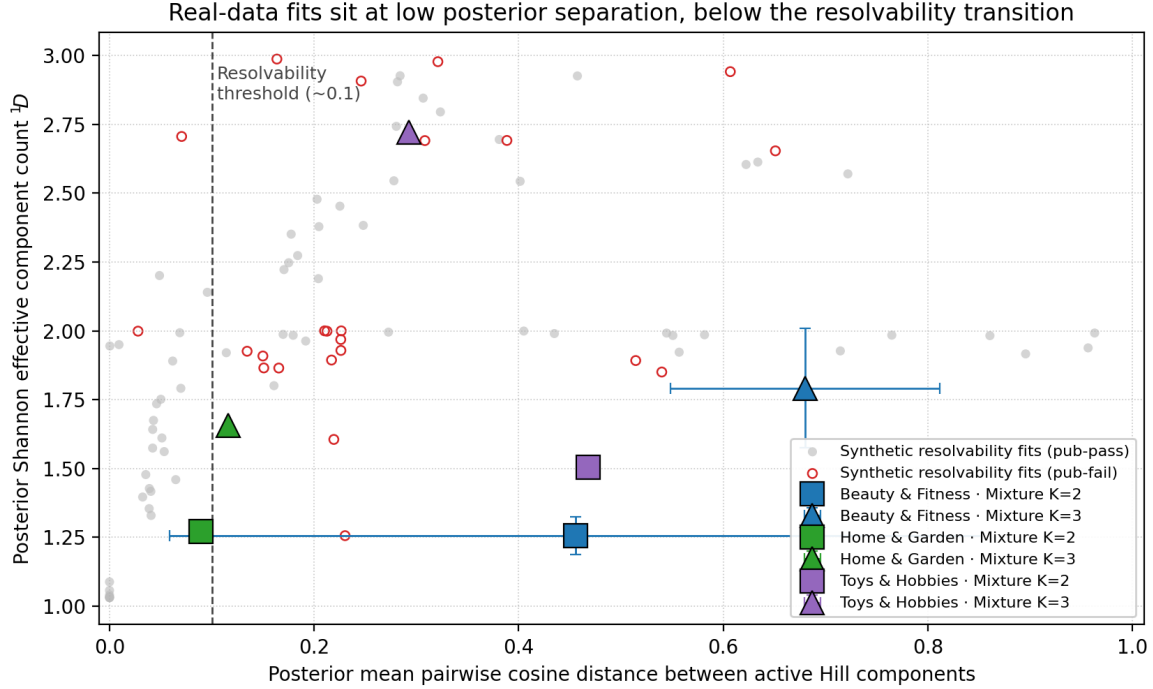


Figure 9: Posterior Shannon effective component count versus posterior mean pairwise cosine distance for the 90 synthetic resolvability fits (grey, with red-outlined fits failing the publication-ready criterion) and the three real organisations averaged across seeds (large coloured markers; error bars are ± 1 s.d. over seeds). The dashed line marks the cosine ≈ 0.10 resolvability transition identified in Section 6.

Two important caveats apply to this overlay. First, the synthetic axis $\bar{d}_{\text{cos}}$ is computed from the *true* component curves while the real axis is computed from *posterior-mean* curves; the latter inherits the chain-mixing problems documented in Table 3 and should be read as a point estimate with substantial uncertainty for the failed fits. Second, the $\bar{d}_{\text{cos}} \approx 0.10$ transition in Section 6 is an empirical, model-dependent boundary rather than a sharp identifiability threshold (e.g. Mixture $K = 3$ on the $K_{\text{true}} = 3$ DGP recovers Shannon count ≈ 2.27 at true cosine 0.058).

Subject to those caveats, the picture is heterogeneous across organisations. *Home & Garden* sits at the resolvability transition: its posterior cosine separation is 0.09 under Mixture $K = 2$ and 0.12 under Mixture $K = 3$, and the Shannon count $\bar{D}_{\text{post}}$ stays at 1.27 and 1.66 respectively. Among the three organisations this fit reaches the convergence threshold most closely (publication-ready $1/3$ for $K = 2$, with the $K = 3$ fit failing only at the strict $\hat{R} \leq 1.01$ bar), so its position on the diagnostic plane is the one we are most confident reporting. *Toys & Hobbies* under Mixture $K = 3$ has a posterior separation of 0.29 and a Shannon count of 2.72, which is inside the resolvable region. However, all three Mixture $K = 3$ *Toys & Hobbies* fits exhibit large numbers of max-tree-depth saturations and tail ESS per chain of 23–34, so the posterior-mean curves driving the 0.29 figure are themselves uncertain; we therefore present this case as consistent with three components rather than as evidence for them. *Beauty & Fitness* occupies the opposite extreme: the highest posterior

separation of any organisation (0.46 for Mixture $K = 2$, 0.68 for Mixture $K = 3$), but every fit fails publication-readiness with $\max \hat{R}$ in the range 2.4–3.6 and tail ESS per chain of order 1–6 for $K = 3$. At that level the posterior summary itself is unreliable; the high posterior separation should be read as a flag for “the model finds widely-separated modes” rather than as a measurement of structure in the data.

The combined reading is that the diagnostic picture is organisation-dependent. Home & Garden lives at the resolvability boundary with reliable enough diagnostics that we can locate it on the plane. Toys & Hobbies is consistent with three components subject to chain-mixing caveats. Beauty & Fitness exhibits strong multi-modality in the posterior coupled with severe chain mixing failures; substantive claims about heterogeneity there should wait for a stronger inference setup or a heavier-tailed likelihood.

8 Discussion: What Worked, What Did Not, What We Did Not Test

We organise the discussion around three honest categories so the reader can separate “the mixture is doing X” from “we did not check Y”. The paper is positioned as a preprint that reports the state of an investigation rather than a finished benchmark; in that spirit, we name failures and untested directions directly.

8.1 What worked

Recovery on the matched DGP. When the synthetic DGP is itself a Hill mixture, the model recovers it: latent-mean nRMSE roughly halves under $K = 2$ and shrinks by $\approx 2.4\times$ under matched $K = 3$, with ELPD gains of +50 and +45 respectively over single-Hill (Section 5).

Predictive density on real data, with caveats. Mixture $K = 3$ has the highest estimated ELPD-LOO in every real organisation, with point gaps +222 to +308 (≈ 2 –4 coarse heuristic SE). These are reported as estimates rather than as established improvements: (i) the proper paired `se_diff` could not be recomputed, (ii) Beauty & Fitness has degraded PSIS reliability, and (iii) *none of the nine real-data Mixture $K = 3$ fits is publication-ready*, so the $K=3$ ELPD numbers come from non-converged inference and should be treated as an exploratory signal rather than a benchmark result (Section 7).

Resolvability and label-invariant diagnostics as working tools. The cosine separation / Shannon count axis (Section 6) gives a concrete vocabulary for placing each real organisation on a diagnostic plane (Section 7.4), and the three-bucket label-invariant diagnostic (Section 3.1) lets us diagnose mixture posteriors in the same workflow as single-Hill.

8.2 What did not transfer

Synthetic–real convergence gap. The benchmark threshold $\hat{R} \leq 1.05$ that 44/45 synthetic fits pass fails on 0/9 Mixture $K = 3$ fits on real data; the stricter publication-ready criterion $\hat{R} \leq 1.01$ passes 33/45 synthetic and 0/9 real-data Mixture $K = 3$ fits. The synthetic DGPs are within the model class with stationary mixing weights, controlled Gaussian noise, and no regime shifts; the real organisations exhibit at least some of these (seasonality, channel-mix changes, outliers). We do not claim to have isolated which of those is the dominant failure cause, only that the gap is real and concentrated on Mixture $K = 3$.

Predictive density does not imply predictive accuracy. Beauty & Fitness ELPD-LOO improves by +222 under Mixture $K = 3$ while holdout MAPE and CRPS *deteriorate*: the likelihood widens around the data more than the mean moves toward it, which is consistent with a Gaussian-mixture absorbing multi-modal residual structure rather than improving point prediction.

Influential observations and non-converging fits. Beauty & Fitness has up to 21 LOO observations with $\hat{k}_i > 0.7$ (Mixture $K = 2$), degrading PSIS reliability. Mixture $K = 3$ on Toys & Hobbies reports tail ESS/chain of 23–34 and on Beauty & Fitness max $\hat{R}$ reaches 2.4–3.6 with tail ESS/chain 1–6. These fits are described and placed on the resolvability plane, but their posterior-mean component summaries are not reported as reliable measurements.

8.3 What we did not test

These are the obvious next questions we did not attempt and have therefore not ruled in or out.

Misspecified synthetic DGPs. All synthetic DGPs in Section 5 and Section 6 are within the fitted model class (Hill family, Gaussian noise, stationary mixing weights, single aggregated channel, $T = 200$). The convergence and recovery numbers should be read as “what happens when the model is correct”, not “what happens when the data look like an industrial MMM problem”. We have not run DGPs with time-varying weights, heavy-tailed noise, non-Hill response shapes, or per-channel mixtures.

Alternative observation models. Two natural variations would be (i) replacing the Gaussian-mixture observation with a sum-mixture $y_t \sim \mathcal{N}(\sum_k \pi_k f_k(s_t), \sigma^2)$, which encodes “one curve composed of K Hill contributions” rather than “one of K Hill curves per observation”, and (ii) replacing the Gaussian noise with Student- t to absorb the Pareto- k -flagged outliers on Beauty & Fitness. Either could change all three real-data conclusions; we did not fit them.

Stronger inference for non-converging fits. Mixture $K = 3$ on Beauty & Fitness and Toys & Hobbies might converge with longer chains, tempering, alternative initialisation, or hand-tuned mass matrices. We did not pursue this; the per-fit summaries on disk are from the configuration in Appendix A and nothing else.

Paired LOO comparisons. We report per-fit PSIS-LOO standard errors but did not retain the pointwise log-likelihood matrices needed to compute the proper paired `se.diff` of Vehtari et al. [2017]. The coarse independent-sum heuristic in Section 7 is not a substitute, and is not a strict upper bound; we report it only for order-of-magnitude calibration.

Causal identification. As Dew et al. [2024] argue, the joint identification of nonlinear and time-varying effects from observational time series is unresolved. Our mixture extension does not address this; it only changes the assumed saturation family. Practitioners using this model for budget decisions should triangulate against geo-experiments or incrementality tests.

9 Scope and Limitations

Beyond the failure modes and untested directions in Section 8, two scope choices limit the generalisability of any positive result above.

Data scope. The real-data evaluation covers three Conjura organisations across three verticals; this is enough to show the synthetic-versus-real diagnostic gap is not a one-off, but not enough to claim cross-organisation generalisation. All experiments aggregate spend across channels into a single x_t ; channel-level mixture-of-Hill, which introduces label switching within each channel, is not evaluated.

Priors and scales. The priors on λ and A depend on empirical spend range and $\text{range}(y)$ respectively. We have not run a prior-sensitivity sweep on the multiplicative constants.

10 Code and Data Availability

All code, configurations, and per-case artifacts are released at

<https://github.com/jpcca/marketing-mix-modelling>

The repository layout:

- `src/hill_mixture_mmm/` – model definitions and inference utilities.
- `src/hill_mixture_mmm/data.py` – synthetic data generator (Section 5).
- `src/hill_mixture_mmm/component_resolvability.py` – resolvability-sweep prior overrides (Section 6; see Appendix B).
- `scripts/run_benchmark.py`, `scripts/run_real_data_validation.py` – synthetic and real-data benchmark drivers.
- `scripts/run_component_resolvability_sweep.py` – resolvability sweep driver.
- `scripts/compute_posterior_separation.py` – posterior cosine separation for Section 7.4.
- `paper/figures/{synthetic,real}/` – per-fit JSON summaries with posterior weights, label-invariant and relabeled diagnostics, and convergence flags.

Random seeds are fixed and recorded in each per-fit JSON. Sampler settings used per fit are stored in `fit_summary`; the resolved prior configuration for each fit (including the resolvability-sweep overrides described below) is determined by the relevant module above rather than per-fit JSON.

11 Conclusion

We presented a Bayesian predictive mixture of Hill saturation functions for MMM, with a label-invariant diagnostic workflow, on a controlled single-aggregated-spend version of the problem. Three findings travel together. (i) Mixture-of-Hill has higher estimated ELPD-LOO than single-Hill on every real organisation, but the gaps are reported as estimates subject to paired-SE and PSIS reliability caveats, and they do not translate into uniform MAPE/CRPS improvements (Beauty & Fitness deteriorates). (ii) The component-resolvability axis gives an empirical, model-dependent transition near cosine ≈ 0.10 , useful as a diagnostic vocabulary rather than a deployable rule. (iii) The benchmark convergence threshold that 44/45 synthetic fits clear fails uniformly for Mixture $K = 3$ on real data; we interpret this synthetic-to-real gap as a directly useful finding for practitioners rather than a weakness to sand away.

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A Inference Hyperparameters

Table 4 reproduces the per-model NUTS settings used throughout the paper. Two chains are run in parallel for every fit; warmup uses NUTS dual averaging with a window size matched to the warmup length. Tree depth is capped at 10, the initialisation strategy is `numpyro.infer.init_to_median`, and mass adaptation uses a diagonal mass matrix.

Table 4: Per-model NUTS configuration. Synthetic and real-data fits differ only in warmup/sampling length; all other settings are shared. The single-Hill synthetic schedule uses 600 warmup/samples on the $K = 1$ and $K = 2$ DGPs and 800 on the $K = 3$ DGP; mixtures use the same schedule across DGPs.

Model	Warmup (synthetic / real)	Samples (synthetic / real)	Chains	Target accept
Single Hill	600–800 / 600	600–800 / 600	2	0.90
Mixture $K = 2$	900 / 900	900 / 900	2	0.95
Mixture $K = 3$	1,600 / 1,600	1,200 / 1,200	2	0.97

B Priors

Table 5 lists the priors actually used by the implementation. Two parameter groups—mixture weights and the saturation parameters (A_k, n_k, λ_k) —use mildly hierarchical or data-anchored constructions rather than independent draws, so we describe them as the implementation specifies them rather than as a single-line distribution. We have not run a formal sensitivity sweep on the constants; this is noted in Section 9.

Table 5: Priors as implemented. Statistics with hats (e.g. $\widehat{\text{range}}(y)$, $\bar{y}$, $\hat{\sigma}_y$) are computed from the training portion only. The mixture weights π are built by stick-breaking from $K - 1$ Beta variables; the per-component log-amplitude and log-slope are drawn from a mildly hierarchical Normal with shared μ and hierarchical σ ; the ordered half-saturations λ_k are anchored to quantiles of the adstocked spend s .

Parameter	Distribution	Notes
Adstock decay α	Beta(2, 2)	shared across components
Stick proportions v_k , $k \leq K-1$	Beta(a, b)	$(a, b) = (3, 1)$ for $K=2$, $(3, 1.5)$ for $K=3$; $\pi_k = v_k \prod_{j < k} (1 - v_j)$
$\mu_{\log A}$ (hier. loc.)	$\mathcal{N}(\log(0.3 \cdot \widehat{\text{range}}(y)), 0.5)$	shared across k
$\sigma_{\log A}$ (hier. scale)	LogNormal(ℓ_A, s_A)	per-experiment ℓ_A, s_A
Per-component log A_k	$\mathcal{N}(\mu_{\log A} + a_k^{\text{anch}}, \sigma_{\log A})$	non-centred; a_k^{anch} optional anchor
$\mu_{\log n}$ (hier. loc.)	$\mathcal{N}(\log 1.5, 0.3)$	shared across k
$\sigma_{\log n}$ (hier. scale)	LogNormal(ℓ_n, s_n)	per-experiment ℓ_n, s_n
Per-component log n_k	$\mathcal{N}(\mu_{\log n} + n_k^{\text{anch}}, \sigma_{\log n})$	non-centred
log λ_1 (anchor)	$\mathcal{N}(\log \text{quantile}_{q_1}(s), s_{\text{anch}})$	q_k equispaced quantiles of s
log $\lambda_{k+1} - \log \lambda_k$	ordered, anchored to quantile $_{q_{k+1}}(s) - \text{quantile}_{q_k}(s)$	strictly positive; scale s_{inc}
Intercept μ_0	$\mathcal{N}(\bar{y}, 2\hat{\sigma}_y)$	data-anchored
Trend β	$\mathcal{N}(0, \hat{\sigma}_y)$	on standardised t ; fallback $\rightarrow 5$
Obs. noise σ	HalfNormal($\hat{\sigma}_y$)	data-anchored

The anchor terms $a_k^{\text{anch}}, n_k^{\text{anch}}$ provide a soft ordering of components in (A, n) space, used to stabilise mixing on weakly identified mixtures. They are controlled by a single scalar ρ_K with implementation defaults $\rho_2 = 0.8$ and $\rho_3 = 0.7$, applied to *all reported Mixture $K = 2$ and Mixture $K = 3$ fits* (both synthetic and real). For $K = 2$ the increment construction is $\log \lambda_2 - \log \lambda_1 = (\log q_2 - \log q_1) + s_{\text{inc}} \cdot |z|$ with $z \sim \mathcal{N}(0, 1)$, ensuring strict positivity; for $K = 3$ the construction is $\log \lambda_{k+1} - \log \lambda_k = \max(\log q_{k+1} - \log q_k + s_{\text{inc}} \cdot z_k, 10^{-3})$, where $z_k \sim \mathcal{N}(0, 1)$ and q_1, q_2, q_3 are

equispaced quantiles of the adstocked spend.

Resolvability-sweep prior override. Table 5 describes the prior configuration used by the synthetic and real-data benchmarks (Sections 5 and 7). The component-resolvability sweep in Section 6 uses a deliberately different prior on the mixture weights and on the half-saturation and amplitude/slope scales, so that the recovery axis is not dominated by the same prior signal that drives the headline benchmarks. The overrides, taken directly from `component_resolvability.py`, are listed in Table 6; all other prior components match Table 5.

Table 6: Prior overrides used by the resolvability sweep of Section 6. All other prior components match Table 5.

Parameter	Mixture $K = 2$ override	Mixture $K = 3$ override
Stick proportions (a, b)	$(0.7, 0.7)$	$(5.0, 2.5)$
λ base scale $s_{\text{anch},0}$	0.30	0.28
λ anchor scale s_{anch}	0.06	0.06
λ increment scale s_{inc}	0.04	0.04
Hier. log- A scale (ℓ_A, s_A)	$(-1.3, 0.20)$	$(-1.5, 0.18)$
Hier. log- n scale (ℓ_n, s_n)	$(-1.8, 0.20)$	$(-1.9, 0.18)$

The exact constants $(\ell_A, s_A, \ell_n, s_n, s_{\text{anch}}, s_{\text{inc}})$ for every experiment are reproduced verbatim in the modules `models.py` and `component_resolvability.py` under `src/hill_mixture_mmm/`, which are the authoritative source.

C Synthetic DGP Parameters

Table 7 summarises the parameter ranges used to generate synthetic data for the three DGPs in Section 5 and the nine profiles used in Section 6. Per-component values are drawn uniformly from the listed ranges and held fixed across the five seeds within each profile.

Table 7: Synthetic DGP parameter ranges. The resolvability profiles fix A_k and n_k and vary only the spacing δ of the half-saturation points λ_k around a fixed centre c .

Parameter	Single ($K = 1$)	Mixture $K = 2$	Mixture $K = 3$
α	0.5	0.5	0.5
λ_k	$[1.0, 2.5]$	$[0.8, 2.8]$	$[0.7, 3.0]$
n_k	$[1.0, 2.0]$	$[1.0, 2.0]$	$[1.0, 2.0]$
A_k	$[80, 120]$	$[80, 130]$	$[80, 140]$
π_k	1	$\{0.6, 0.4\}$	$\{0.5, 0.3, 0.2\}$
σ	5.0	5.0	5.0
T	200	200	200
Train ratio	0.75	0.75	0.75